

Khadeeja

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PROFESSIONAL SUMMARY

Data Science graduate student at Aligarh Muslim University with hands-on experience building machine learning systems, deep learning models, NLP pipelines, and GenAI agents. Comfortable working across the full stack from raw data to deployed products. A dedicated, focused, and driven individual who leads projects end-to-end with a strong bias for execution.

EDUCATION

Master of Science (Data Science) Aligarh, India
Aligarh Muslim University 2024 – 2026

TECHNICAL SKILLS

Languages & Tools: Python, R, SQL, MySQL, Git, GitHub, VS Code, Google Colab, Excel, LaTeX
Machine Learning & Deep Learning: Scikit-Learn, TensorFlow, Keras, XGBoost, CNN, LSTM, GRU, ResNet-50, ViT, GNN, Multimodal Deep Learning, PCA, SMOTE, Cross-Validation
NLP, LLMs & GenAI: BERT, Hugging Face Transformers, LangChain, LangGraph, FAISS, RAG Pipelines, Prompt Engineering, Sentence Transformers, Vector Databases, Multi-Agent Systems, Vision-Language Models (VLMs)
Deployment & Interfaces: FastAPI, Flask, Streamlit, Gradio, Hugging Face Spaces
Data Analysis & Visualisation: Pandas, NumPy, Matplotlib, Seaborn, Tableau, Power BI, Feature Engineering, ETL Pipelines

EXPERIENCE

Research Intern June 2026 – Present
Insight Centre for Data Analytics, University of Galway, Ireland Supervisor: Prof. Saeed H. Alsamhi

- Developing a multi-agent AI framework for skin disease diagnosis using Vision-Language Models, RAG, and memory-based reasoning.
- Researching multimodal AI, agentic systems, and dermatology foundation models for healthcare applications.

PROJECTS

Multimodal Fake Review Detection System 2026
BERT | ViT | TensorFlow | Cross Modal Attention | GNN | Hugging Face Spaces

- Developed end-to-end multimodal deep learning system using BERT (text) + Vision Transformer (image) achieving 99.72% F1-score on 12,254 samples; implemented novel Cross-Modal Attention architecture for detecting text-image inconsistencies in fraudulent reviews.
- Engineered feature extraction pipeline: 768-dim BERT embeddings + 768-dim ViT embeddings with attention-based fusion; conducted ablation study and SHAP explainability analysis revealing equal importance of text (48.7%) and image (51.3%) modalities.
- Benchmarked 7 classification models (Logistic Regression, XGBoost, LightGBM, FFNN, CMA, GNN, Naive Bayes) using Tensorflow; implemented data preprocessing including mean imputation, KNN imputation, and stratified sampling.
- Deployed as interactive web application on Hugging Face Spaces for real-time inference; users can upload review text and images to get authenticity predictions.

Conversational AI Lead Generation Agent | AutoStream 2026
Python | LangChain | LangGraph | FAISS | Sentence Transformers | Groq API | Flask

- Engineered a LangGraph agentic workflow routing 3 intent types (greeting, inquiry, purchase) across multi-turn conversations automatically.
- Engineered a FAISS-based RAG pipeline with Sentence Transformers to retrieve accurate answers from a curated knowledge base, eliminating hallucinations.

- Implemented a conditional lead capture mechanism recording Name, Email, and Platform only after confirming buying intent, storing data in CSV.
- Deployed Flask chat interface with voice responses; conversation state persists across 10+ dialogue turns via LangGraph state management.

Parkinson's Disease Detection | Multi-Model ML Pipeline

2025

Python | XGBoost | ANN | TensorFlow/Keras | SMOTE

- Benchmarked 7 classifiers - Logistic Regression, Naive Bayes, KNN, Random Forest, SVM, XGBoost, Custom ANN - on 195 biomedical voice samples.
- Applied SMOTE post-stratified split to resolve class imbalance (147 PD vs 48 healthy) without leaking synthetic samples into evaluation.
- Custom ANN with L2 regularisation and early stopping achieved 97.44% accuracy, 100% precision, 96.55% recall, F1 0.98, ROC-AUC 0.99.
- Reduced XGBoost overfitting via manual regularisation - max_depth, subsample, colsample_bytree, improving test accuracy to 87.18%.

Impact of Social Media on Student Mental Health | Statistical Study

2024

R | Hypothesis Testing | Stratified Sampling | Survey Design | Google Forms

- Conducted a stratified random sampling survey of 200 students at Women's College AMU on social media and mental health.
- Performed hypothesis testing in R; found 58.4% reported negative academic impact, 53.7% experienced loneliness, 56.7% had mood changes.
- Validated effects on sleep quality and focus through chi-square and t-tests, producing a peer-reviewable statistical report.

ACHIEVEMENTS & CERTIFICATIONS

IBM Data Analysis with Python — Cognitive Class (IBM)

IBM Machine Learning with Python — Cognitive Class (IBM)

Python for Machine Learning — Kaggle

Python Fundamentals Meets AI Workshop — Marine Technology Society, AMU

Deep Dive into AI Workshop — Marine Technology Society, AMU